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Mo Global Oil Industry Monthly Report

Oct 30, 2026 · Inaugural Issue (Issue 1)

Part I • Upstream Panorama: 99 producing countries, oilfields, rigs, oilfield services, infrastructure, projects under construction and blocks on offer
Part II • Downstream Panorama: transport fuels, petrochemicals and chemical fibres, fertilisers and pesticides, pharmaceuticals, food and industry output value

84.38	99	1,572.5	$107.03
Global crude output (million bbl/d, 2025, EIA basis, +3.0% y/y)	Countries with crude output in 2025 (EIA list), all covered in this issue	Global proved reserves (billion bbl, OPEC basis, excl. Canadian oil sands)	Brent spot price (USD/bbl, Nov 3, 2026, EIA)

This report is the monthly publication of the Mo series. Data in this issue are as of November 8, 2026.

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Key Points and Contents

Key Conclusions

- **2025 was a record year.**Global crude output was 84.38 million bbl/d (+3.0% y/y) and total liquids output 106.29 million bbl/d; the United States led at 13.59 million bbl/d, about 70% of Russia (9.89) and Saudi Arabia (9.56) combined.
- **Treat the 2026 numbers with caution.**After military action began on February 28, 2026, flows through the Strait of Hormuz fell from 21.6 million bbl/d in Q4 2025 to 4.9 million bbl/d in Q2 2026, and Brent spiked to $138.21 on April 7. All 2026 figures in this issue are wartime estimates.
- **New supply is coming from South America and Africa.**Guyana, Brazil, Uganda, Suriname and Angola are the main start-up story for 2025–2028; the list of blocks currently on offer is in Chapter 9.
- **Downstream demand has not collapsed.**Transport accounts for about 58% of oil consumption and petrochemical feedstock for 15.8%, the latter driving almost all net demand growth since 2019 — crude's value is anchored in real downstream consumption.

Contents

Part I — Upstream: Oil Production	Page	Part II — Downstream: The Oil Industry	Page
1. Global Overview: Four Definitions First	3	14. Oil Consumption Structure	34
2. Producing Countries at a Glance (with the Full 99-Country Table)	5	15. Transport Fuels	36
3. Proved Reserves and the Reserves-to-Production Ratio	10	16. Petrochemicals and Chemical Fibres	37
4. Oil Wells and Rigs	12	17. Fertilisers and Pesticides	39
5. World-Class Giant Oilfields	14	18. Pharmaceuticals and Daily-Use Chemicals	41
6. Oilfield Services (OFS)	16	19. Food and Packaging	43
7. Oilfield Infrastructure: Pipelines, Terminals and Refineries	18	20. Industry Output Value in Summary	44
8. Oilfields Under Construction and Development	20	21. How Mo Connects with the Downstream	46
9. Oilfields and Blocks Seeking Investors	24	Appendix I — Data Notes and Key Sources	48
10. Institutions and the Landscape: OPEC, OPEC+ and the IEA	26	Appendix II — Conflicting Definitions and the No-Public-Data List	49
11. Supply, Demand and Trade	28		
12. Trends, Outlook and the Competitive Landscape	30		
13. Special Focus: Chad and Canada	32		

How to read this report: Output is uniformly in million bbl/d; reserves in billion bbl. When different institutions give different numbers for the same thing, it is a difference of measurement basis, not an error — Chapter 1 sets out the four bases first, and every table below states its basis and source. Where no figure can be found, this report writes “No public data” rather than estimating or inventing one.

Part I — Upstream: Oil Production | 1. Global Overview

Four Definitions First, or the Numbers Won't Add Up

Ask "how much oil does the world produce in a year?" and you get four answers: 74.85, 84.38, 100.59 and 106.29. They are not measuring the same thing: the OPEC basis is the narrowest, counting only crude within its statistical scope; the EIA crude basis includes lease condensate; the Energy Institute (EI) adds natural gas liquids (NGL); and the EIA total liquids basis is the widest, adding biofuels and refinery gain. Production rankings in this report always use the EIA crude basis, and any other basis is flagged where it is used.

Indicator	2024	2025	Source and Basis
Global crude output (incl. lease condensate)	81.95	**84.38** (+3.0%)	EIA International Energy Statistics
Global total liquids output (crude + NGL + biofuels + refinery gain)	approx. 103.1	**106.29**	EIA STEO (October 2026)
Global oil output (EI basis: crude + condensate + NGL)	approx. 97.2	**100.59** (+3.5%)	EI Statistical Review of World Energy 2026
Global crude output (OPEC basis)	approx. 72.61	74.85	OPEC Annual Statistical Bulletin 2026; conflicts with the EIA basis, see Appendix II
Global oil consumption / demand	102.81	104.40 (EIA liquids consumption) / 105.15 (OPEC demand)	EIA STEO; OPEC Annual Statistical Bulletin
Global proved reserves (billion bbl, year-end)	1,570.3	**1,572.46**	OPEC basis, excl. Canadian oil sands
Global proved reserves (billion bbl, Eni basis)	approx. 1,761	**1,802.5**	Eni World Energy Review 2026, incl. oil sands
OPEC+ effective spare capacity	approx. 5.87	approx. 4.02 (December 2025, IEA)	IEA Oil Market Report; 2026 cannot be measured reliably because of the Strait disruption

Units are million bbl/d unless stated otherwise. The two reserves figures differ by about 230 billion bbl, almost entirely because of whether Canadian oil sands count as "proved reserves": the OPEC basis credits Canada with 5.08 billion bbl, the Eni basis with 201.3 billion bbl. When reading the reserves table (Chapter 3), check the basis column first.

How to read 2026. The EIA STEO (October 2026) forecasts full-year 2026 total liquids output of 101.09 million bbl/d and consumption of 102.43 million bbl/d, both below 2025 — not a demand collapse, but war-driven shutdowns and transport disruption. The forecast is revised monthly; this report cites it only as a point-in-time estimate and does not compare it on the same basis with 2025 actuals.

Oil Prices: Down in 2025, Pushed Up by War in 2026

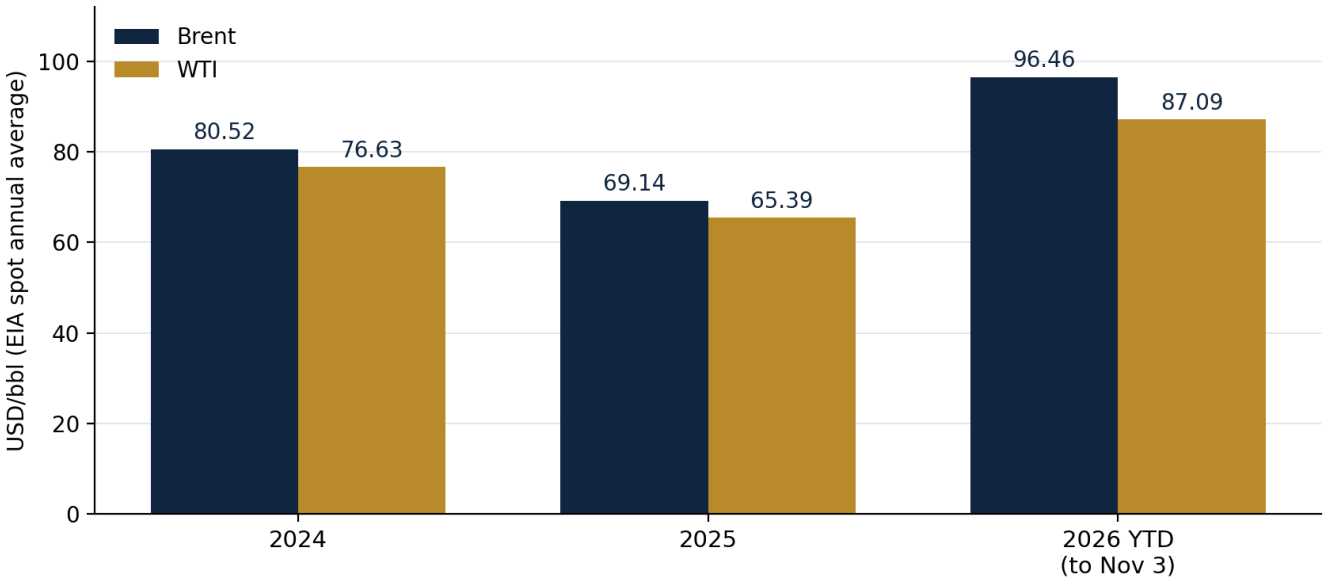


Figure 1 • Brent and WTI annual average spot prices (USD/bbl). 2026 is the year-to-date average to November 3. Source: EIA spot prices.

Price Point	Brent	WTI	Notes
2024 average	80.52	76.63	EIA spot
2025 average	69.14	65.39	About −14% y/y, mainly pre-war oversupply
2026 YTD average (to Nov 3)	96.46	87.09	EIA spot
Peak on April 7, 2026	138.21	—	Highest spot price after the Hormuz disruption
September 2026 average	approx. 114	—	Still high after a partial recovery in Strait flows
Spot on Nov 3, 2026	**107.03**	**99.64**	EIA daily spot; Brent futures $98.32 on Nov 6
Full-year 2026 forecast (EIA STEO)	96.32	88.21	Brent forecast to fall back to about $84 in 2027

This round of price rises is a supply shock, not demand-led: output is stuck behind the Strait, while the IEA revised demand down by 2.5 million bbl/d for the full year in its September 2026 report. For oilfield assets this means two things: first, high prices amplify current sales revenue from producing fields; second, the IEA also notes that upstream investment has not followed prices up (see Chapter 12) — if high prices do not last, owners value arrangements that lock in long-term cash flow all the more.

Reliability of wartime data. The EIA has stated explicitly that tanker-tracking data since late February 2026 is "particularly unreliable", and Gulf output and exports are estimates revised month by month. Every 2026 figure in this report carries its date and source, and later issues will update them with revised values.

Part I — Upstream: Oil Production | 2. Producing Countries at a Glance

The Top Five Produce Half; the Top Ten, 72.1%

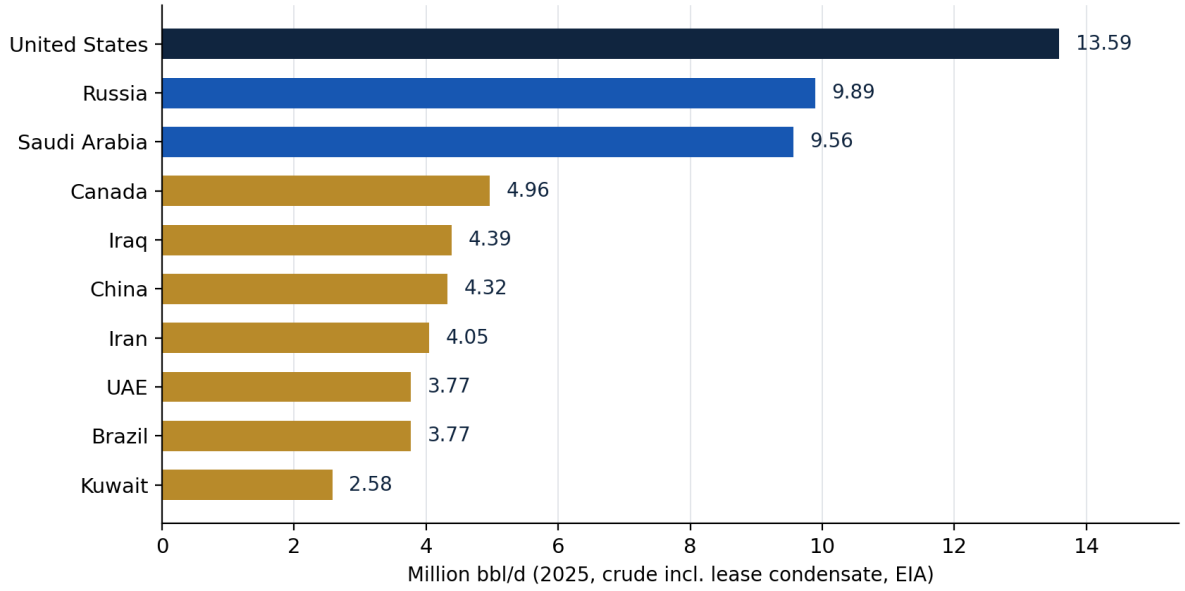


Figure 2 • Top ten crude producers in 2025 (million bbl/d). Source: EIA International Energy Statistics.

Rank	Country	2025 Output	Global Share	Y/Y	Approx. million t/yr*
1	United States	13.59	16.1%	+2.7%	677
2	Russia	9.89	11.7%	-0.1%	492
3	Saudi Arabia	9.56	11.3%	+3.5%	476
4	Canada	4.96	5.9%	+3.9%	247
5	Iraq	4.39	5.2%	-1.6%	219
6	China	4.32	5.1%	+2.1%	215
7	Iran	4.05	4.8%	+2.7%	202
8	UAE	3.77	4.5%	+2.7%	188
9	Brazil	3.77	4.5%	+12.3%	188
10	Kuwait	2.58	3.1%	+1.2%	129

*Million tonnes per year is an approximate conversion at 7.33 bbl/t; crude densities range from 7.0 to 7.9 bbl/t, so the error is about ±5%. These are not the source's own figures. Shares are calculated against global output of 84.38 million bbl/d.

Ranks 11 to 30, with the Five-Year Trend

Rank	Country	2025	Share	Y/Y	Rank	Country	2025	Share	Y/Y
11	Kazakhstan	2.05	2.4%	+13.9%	21	Argentina	0.79	0.9%	+12.9%
12	Norway	1.86	2.2%	+3.9%	22	Guyana	0.75	0.9%	+21.0%
13	Mexico	1.73	2.0%	-6.3%	23	Colombia	0.75	0.9%	-2.6%
14	Nigeria	1.61	1.9%	+8.1%	24	United Kingdom	0.62	0.7%	+5.1%
15	Libya	1.36	1.6%	+17.2%	25	Indonesia	0.58	0.7%	0.0%
16	Qatar	1.30	1.5%	+0.8%	26	India	0.58	0.7%	-1.7%
17	Algeria	1.14	1.4%	+1.8%	27	Azerbaijan	0.56	0.7%	-5.1%
18	Angola	1.03	1.2%	-9.6%	28	Malaysia	0.51	0.6%	+6.3%
19	Oman	1.00	1.2%	+1.0%	29	Egypt	0.50	0.6%	-7.4%
20	Venezuela	0.97	1.1%	+12.8%	30	Ecuador	0.44	0.5%	-8.3%

Five-Year Trend (2020 → 2025, million bbl/d)

Country	2020	2021	2022	2023	2024	2025	Change over Five Years
World	75.82	—	—	—	81.95	84.38	+11.3%
United States	11.31	11.31	12.00	12.94	13.23	13.59	+20%
Iran	2.51	2.95	3.14	3.52	3.95	4.05	+61%, recovery after sanctions
Guyana	0.07	0.12	0.29	0.39	0.62	0.75	approx. 10.7×
Argentina	0.48	0.51	0.58	0.64	0.70	0.79	+65%, Vaca Muerta shale
Canada	4.18	4.44	4.54	4.59	4.78	4.96	+19%, steady year-by-year growth
United Kingdom	0.95	0.81	0.74	0.66	0.59	0.62	−35%, North Sea decline
Angola	1.24	1.11	1.15	1.14	1.14	1.03	-17%

The fastest growers in 2025 were Guyana (+21.0%), Libya (+17.2%), Kazakhstan (+13.9%), Argentina (+12.9%), Venezuela (+12.8%) and Brazil (+12.3%); the biggest fallers were Angola (−9.6%), Ecuador (−8.3%), Egypt and Australia (−7.4% each) and Mexico (−6.3%).

Full Table of the 99 Producing Countries (I): North America, South & Central America and the Caribbean, Europe, Middle East

Basis: crude incl. lease condensate, 2025 daily average, in thousand bbl/d. Source: EIA International Energy Statistics, compiled by AtlasBig (updated November 7, 2026); world total 84,463 thousand bbl/d across 99 countries. Russia is transcontinental and is counted, by convention, in the Europe group only once; Türkiye is counted in the Middle East group, and Kazakhstan and Azerbaijan in the Asia-Pacific group (see next page). Regional subtotals depend on the grouping and should not be cited as standalone statistics.

North America (6 countries)

Country	Thousand bbl/d	Country	Thousand bbl/d	Country	Thousand bbl/d
United States	13,662	Canada	4,964	Mexico	1,726
Cuba	25.5	Guatemala	5.4	Belize	0.79

South & Central America and the Caribbean (12 countries)

Country	Thousand bbl/d	Country	Thousand bbl/d	Country	Thousand bbl/d
Brazil	3,769	Venezuela	973	Argentina	794
Guyana	747	Colombia	746	Ecuador	441
Trinidad and Tobago	51.2	Peru	44.2	Bolivia	17.7
Suriname	12.1	Chile	2.0	Barbados	1.0

Europe (23 countries)

Country	Thousand bbl/d	Country	Thousand bbl/d	Country	Thousand bbl/d
Russia	9,885	Norway	1,858	United Kingdom	616
Italy	83.7	Denmark	71.8	Romania	51.5
Germany	31.6	Ukraine	31.2	Belarus	25.7
Hungary	23.4	Netherlands	21.5	Poland	16.5
Albania	12.2	Serbia	11.5	France	10.2
Croatia	9.0	Austria	8.8	Greece	1.5
Czechia	1.3	Bulgaria	1.0	Lithuania	0.7
Spain	0.03	Slovakia	0.02		

Middle East (13 countries)

Country	Thousand bbl/d	Country	Thousand bbl/d	Country	Thousand bbl/d
Saudi Arabia	9,549	Iraq	4,388	Iran	4,051
UAE	3,771	Kuwait	2,584	Qatar	1,297
Oman	1,002	Bahrain	176	Türkiye	125
Syria	75.2	Israel	19.5	Yemen	15.0

Jordan	0.02

Full Table of the 99 Producing Countries (II): Africa, Asia-Pacific

Africa (20 countries)

Country	Thousand bbl/d	Country	Thousand bbl/d	Country	Thousand bbl/d
Nigeria	1,605	Libya	1,363	Algeria	1,143
Angola	1,031	Egypt	505	Congo (Brazzaville)	241
Gabon	239	Ghana	183	**Chad**	**126**
South Sudan	113	Niger	101	Senegal	100
Equatorial Guinea	76.4	Cameroon	59.2	Côte d'Ivoire	55.1
Sudan	30.0	Tunisia	25.3	Congo (Kinshasa)	16.0
South Africa	0.37	Morocco	0.03		

Asia-Pacific (incl. Central Asia and Oceania, 25 countries)

Country	Thousand bbl/d	Country	Thousand bbl/d	Country	Thousand bbl/d
China	4,325	Kazakhstan	2,047	Indonesia	582
India	581	Azerbaijan	571	Malaysia	513
Australia	246	Turkmenistan	191	Vietnam	165
Thailand	162	Brunei	101	Pakistan	57.5
Papua New Guinea	31.5	Uzbekistan	29.6	Mongolia	13.7
New Zealand	7.9	Myanmar	5.6	Kyrgyzstan	5.5
Japan	3.1	Bangladesh	3.0	Timor-Leste	2.0
Philippines	0.5	Tajikistan	0.32	Georgia	0.22
Taiwan, China	0.2				

Reconciliation: 6 (North America) + 12 (South & Central America and the Caribbean) + 23 (Europe) + 13 (Middle East) + 20 (Africa) + 25 (Asia-Pacific) = 99 countries, matching the EIA list total. Every country in the list has a 2025 figure; the smallest are Slovakia and Jordan at 0.02 thousand bbl/d each (about 20 bbl/d). Namibia, Uganda and Suriname (offshore) had no commercial crude output in 2025 and are not in this table; their project progress is in Chapter 8. A separate EIA compilation lists 101 countries (including Slovenia at zero); the difference is zero-value entries and does not affect the conclusions.

This full table draws a boundary: the countries with real scope for large-scale oilfield-asset cooperation are the roughly 30 whose output exceeds 100,000 bbl/d and which have mechanisms for foreign cooperation and licensing rounds; the licensing-round list in Chapter 9 and the tracking focus of future issues both fall within this set.

Part I — Upstream: Oil Production | 3. Proved Reserves

The Top Three Hold Nearly Half the Reserves but Produce Only About a Sixth

Venezuela, Saudi Arabia and Iran together hold 49.6% of global proved reserves (OPEC basis) but accounted for only about 17% of 2025 output. The reserves-to-production (R/P) ratio is a static measure: how many years of production remain if output stays flat and no new reserves are added. It is not a forecast of exhaustion — the US R/P ratio has hovered around 10 years for years, yet shale keeps replenishing reserves and output keeps hitting new highs.

Rank	Country	Reserves (billion bbl)	Share	R/P Ratio (years)*	Notes
1	Venezuela	303.70	19.3%	approx. 858	The extreme R/P ratio reflects very low output
2	Saudi Arabia	267.20	17.0%	approx. 77	
3	Iran	208.60	13.3%	approx. 141	Reserves figure unchanged since 2019
4	Iraq	140.00	8.9%	approx. 87	Revised down 5.0 billion bbl in 2025
5	UAE	120.00	7.6%	approx. 87	Revised up 7.0 billion bbl in 2025
6	Kuwait	101.50	6.5%	approx. 108	Figure unchanged for years
7	Russia	80.00	5.1%	approx. 22	Eni basis: 107.8
8	Libya	48.36	3.1%	approx. 97	
9	United States	41.91	2.7%	approx. 8.4	EIA basis 45.95 (end-2024), about 9.3 years
10	Nigeria	37.01	2.4%	approx. 63	
11	China	34.47	2.2%	approx. 22	Revised up 6.3 in 2025, the year's largest upward revision
12	Kazakhstan	30.00	1.9%	approx. 40	Figure unchanged since 2017
13	Qatar	25.24	1.6%	approx. 53	Figure unchanged since 2017
14	Brazil	16.84	1.1%	approx. 12	Brazil ANP basis: 17.49
15	Algeria	12.20	0.8%	approx. 29	
16-20	Ecuador 8.27 • Azerbaijan 7.00 • Norway 5.92 • Mexico 5.14 • Canada 5.08	—	—	—	Canada is on the excl. oil sands basis; on the Eni basis Canada is 201.3, R/P ratio about 111 years
—	**World (OPEC basis)**	**1,572.46**	100%	approx. 51	Eni basis world total 1,802.5; OPEC members combined 1,243.49 (79.1%)

*R/P ratios are calculated by this report: reserves ÷ (2025 crude output × 0.365). Source: OPEC Annual Statistical Bulletin 2026, Table 3.1 (end-2025). Note: several OPEC members' reserves are self-reported figures, unrevised for years and not independently audited — cite them with this qualification.

Only the US Has a Complete Answer on Well Counts; Other Countries Have No Public Data

This chapter needs a clear statement up front: there is no unified global count of producing oil wells. Apart from the US EIA, which publishes one every year, other countries have only a single old snapshot from the Oil & Gas Journal (OGJ) in 2014. This report lists the old figures with their year stated, and writes “No public data” where nothing can be found — old numbers are not passed off as current ones.

Country	Producing Wells	Year and Basis
United States	**918,481**	2024, oil and gas wells combined (EIA); 2014 peak 1,031,161 wells; horizontal wells rose from 10% in 2014 to 22% in 2024; about 78% of wells produce no more than 15 boe/d, and the median per-well output is very low
Russia	124,581	2014 (OGJ snapshot, out of date, for scale only)
Canada	82,556	2014 (as above)
China	72,691	2014 (as above)
Venezuela	14,651	2014 (as above)
Saudi Arabia	2,895	2014 (as above); per-well output of about 3,402 bbl/d implied by that year's production, the highest in the world — a small number of very high-output wells is typical of the Gulf producers
Iraq, Iran, UAE, Kuwait, Brazil, Norway, Kazakhstan, Nigeria, Chad and other remaining countries	**No public data**	National oil companies do not publish producing-well counts on this basis; this report does not estimate them

Rig Counts (Baker Hughes)

Scope	Latest Figure	Notes
World (monthly average)	1,919 (September 2026); 1,933 (October)	1,812 in September 2025
International (excl. US and Canada)	1,122 (September 2026)	1,084 in September 2025
United States (weekly)	604 (week ending November 6, 2026), of which 463 oil rigs	About +57 y/y; Texas 275, New Mexico 99 (October 30)
Canada (weekly)	223 (November 6, 2026)	226 on October 30, of which 154 oil rigs
Saudi Arabia	Total rigs about 262–269 (2026)	Conflicting bases for oil rigs: one says 66, another 118; different bases, shown side by side, not merged

Regional distribution (October 2025, somewhat dated)	North America 741 • Middle East 505 • Asia-Pacific 204 • Latin America 129 • Europe 124 • Africa 97	An updated country breakdown sits behind Baker Hughes' paywall and could not be obtained from public sources

The onshore/offshore split of output likewise has no single official basis: the IEA estimates offshore at about 30% of global crude output, while the Rystad basis puts offshore oil and gas combined at about 50 million boe/d, roughly half of total oil and gas output — different units that cannot be substituted for each other. About 95% of Brazil's output is offshore (pre-salt); Norway, the UK, Guyana and Angola are almost entirely offshore; the US Gulf of Mexico produces about 1.8 million bbl/d offshore, about 13% of US crude output.

Part I — Upstream: Oil Production | 5. World-Class Giant Oilfields

The Giant-Field List: First, a Caveat — Field-Level Output Has No Audited Annual Report

The figures below are publicly reported capacity or recent output approximations; "capacity" and "actual output" are often used interchangeably in the original reporting. Ghawar alone is more than twice Kazakhstan's entire national output.

Field / Basin	Country	Output or Capacity (million bbl/d)	Onshore/Offshore	Operator
Permian Basin (a basin, not a single field)	United States	6.6 (2025 average, EIA)	Onshore shale	Many independents and majors
Ghawar	Saudi Arabia	3.8–5.0 (capacity, commonly cited as about 4.5)	Onshore	Saudi Aramco
Burgan	Kuwait	approx. 1.7	Onshore	Kuwait Oil Company
Rumaila	Iraq	approx. 1.5–1.6	Onshore	Basra Oil Company / BP–CNPC service contract
Safaniya	Saudi Arabia	approx. 1.2–1.5 (capacity; current output unverified)	Offshore, the world's largest offshore field	Saudi Aramco
Ahvaz	Iran	approx. 1.2	Onshore	National Iranian Oil Company
Khurais / Shaybah	Saudi Arabia	approx. 1.2 / approx. 1.0 (both capacity)	Onshore	Saudi Aramco
West Qurna (Phases 1 and 2 combined)	Iraq	approx. 0.8–1.0 (per-field figures unverified)	Onshore	CNPC (Phase 1, succeeding ExxonMobil); LUKOIL (Phase 2)
Upper Zakum	UAE	approx. 0.9–1.0 (target capacity)	Offshore	ADNOC Offshore / ExxonMobil / INPEX
Tengiz	Kazakhstan	0.93 (May 2025, after expansion)	Onshore	Tengizchevroil (Chevron 50%, ExxonMobil 25%, KazMunayGas 20%)
Búzios	Brazil	0.80 (February 2025 record)	Offshore pre-salt	Petrobras
Lula / Tupi	Brazil	Peak approx. 0.7–1.0 (2025 per-field figures unverified)	Offshore pre-salt	Petrobras
Daqing	China	approx. 0.6	Onshore	CNPC
Johan Sverdrup	Norway	approx. 0.66–0.75 (plateau)	Offshore	Equinor
Kashagan	Kazakhstan	approx. 0.41–0.45	Offshore (Caspian Sea)	North Caspian Operating Company (Eni, Shell, TotalEnergies and others)

Samotlor	Russia	approx. 0.36	Onshore (West Siberia)	Rosneft
Stabroek Block (Liza, Payara, Yellowtail)	Guyana	2025 average 0.75, over 0.90 by year-end	Offshore deepwater FPSO	ExxonMobil (operator), Hess/ Chevron, CNOOC
Prudhoe Bay	United States (Alaska)	Peak approx. 1.5 (1980s); current output unverified	Onshore (Arctic)	Hilcorp (succeeding BP since 2020)
Cantarell	Mexico	approx. 0.16	Offshore	Pemex

Oilfield Services: All Three Giants' Revenues Fell in 2025; Subsea Is the Only Growth Pocket

There is no agreed single size for the oilfield services market: on the mainstream (services-only) basis it was about $119–144 billion in 2024–2025, while some firms that also count EPC contractors and drilling contractors give $320–350 billion. This report uses the mainstream range; the wide basis appears only in footnotes, and no average is taken.

Firm's Basis	2024	2025	Forecast
Mordor Intelligence	$118.96bn	$126.32bn	$167.69bn by 2030, +5.83% p.a.
WiseGuy Reports	$140.4bn	$143.6bn	$180bn by 2035
Verified Market Research	$134.01bn (year labels inconsistent)		$169.76bn by 2033

Company	2024 Revenue	2025 Revenue	Main Business and Notes
SLB (Schlumberger)	$36.29bn	$35.71bn (−1.6%)	Digital, reservoir, well construction and production systems; about 109,000 employees, the industry's largest
Baker Hughes	$27.83bn	$27.73bn (−0.3%)	Oilfield services & equipment about 51.7%, industrial & energy technology about 48.3%; 72% of revenue from outside the US
Halliburton	$22.94bn	$22.18bn (−3.3%)	Completion & production, drilling & evaluation; strongest in North American shale fracturing
Saipem (Italy)	€14.549bn	€15.497bn (+6.5%)	Offshore engineering, energy carriers, offshore drilling; order backlog €31.5bn
TechnipFMC	$9.08bn	$9.93bn (+9.4%)	Subsea new orders $10.1bn in 2025, order backlog $16.6bn
NOV	$8.9bn	$8.7bn (−1.4%)	Rig equipment and wellbore technologies; more equipment than service
Weatherford	$5.51bn	$4.92bn (−10.8%)	Drilling evaluation, well construction and completion, production and intervention; the steepest fall
Petrofac		**Unverified**	UK engineering services company, in restructuring; no reliable revenue figure obtained, not cited

Units are USD billion unless stated otherwise. Three structural trends: first, high concentration — on a rough calculation from listed companies' 2025 revenues, SLB, Baker Hughes and Halliburton together account for about 52% of comparable listed-company revenue (this share comes from a secondary compilation, not audited market research, and is indicative of scale only); second, North America weakened, and international business, which held up growth in 2024, also turned weaker in 2025; third, subsea is the strongest segment, with TechnipFMC's cumulative subsea orders over three years at about

$30 billion. The logging sub-market is even more concentrated: SLB 28.5%, Halliburton 22.0%, Baker Hughes 18.2%, 68.7% combined.

The OFS giants' customers are oilfield owners, and their revenues move with owners' capital spending; the 2025 fall across all three giants corroborates the upstream investment contraction (Chapter 12). For small and mid-sized field owners, the availability and pricing of oilfield services — especially drilling, completion and stimulation — are hard variables in cash-flow calculations, as important as reserves and production data.

Part I — Upstream: Oil Production | 7. Oilfield Infrastructure: Pipelines, Terminals and Refineries

China Leads in Refining Capacity; the US Leads in Actual Throughput

Rank	Country	Refining Capacity (thousand bbl/d)	Throughput (thousand bbl/d)	Utilisation Rate*
1	China	18,829	14,747	78.3%
2	United States	18,172	16,371	90.1%
3	Russia	6,718	4,900	72.9%
4	India	4,924	5,468	111.1% †
5	South Korea	3,570	2,812	78.8%
6	Saudi Arabia	3,289	2,741	83.3%
7	Japan	3,110	2,367	76.1%
8	Iran	2,460	2,019	82.1%
9	Brazil	2,296	1,977	86.1%
10	Germany	1,951	1,642	84.2%
11	Canada	1,943	1,737	89.4%
12	Mexico	1,847	1,017	55.1%
—	**World**	**103,756**	**83,102**	80.1%

Source: EI Statistical Review, compiled by Energy Factbook (updated August 2026), 2025. *Utilisation rate = throughput ÷ year-end capacity; † India above 100% reflects a statistical timing difference (source's own note). Regional capacity: Asia-Pacific 38,025, North America 21,962, Europe 14,389, Middle East 11,936 (thousand bbl/d). Other top-20 countries: Italy 1,673, Spain 1,532, Kuwait 1,430, Iraq 1,389, Venezuela 1,303 (utilisation only 21.3%), Singapore 1,289, Netherlands 1,260, Indonesia 1,259.

Major Crude Export Pipelines

Pipeline	Route	Capacity (million bbl/d)	Status
Druzhba (Friendship) Pipeline	Russia → Central and Eastern Europe	1.2–1.4 (peak)	In service since 1964, about 5,500 km
ESPO	East Siberia → China / Kozmino Port	approx. 1.0 (design up to 1.6)	4,857 km
CPC Caspian Pipeline	Tengiz → Novorossiysk	approx. 1.35	About 80% of Kazakhstan's exports; interrupted several times
BTC	Baku → Ceyhan (Türkiye)	Design 1.2, actual about 0.6	1,768 km
Trans Mountain Pipeline TMX (after expansion)	Edmonton → Burnaby (Canada)	0.89	In service since May 2024; target about 1.19 by end-2028
Chad–Cameroon Pipeline	Doba Basin → Kribi (Cameroon)	Design about 0.225	1,070 km, Chad's only export route, see Chapter 13
EACOP (under construction)	Uganda → Tanga (Tanzania)	approx. 0.216 (another source says 0.246; conflict flagged)	1,443 km, the world's longest heated crude pipeline, 92.7% complete (September 2026)

Refinery Openings and Closures

New builds and ramp-ups: Nigeria's Dangote refinery, 650,000 bbl/d, utilisation about 60–65%, planned expansion to 1.4 million bbl/d; Mexico's Dos Bocas, 340,000 bbl/d. Closures: Kpler counts about 800,000–900,000 bbl/d of permanent closures in Europe and the US in 2025–2026, and the IEA expects the closure wave to continue. Major export terminals include Kozmino, Novorossiysk, Ceyhan, Ras Tanura, Fujairah, LOOP and Rotterdam; there is no unified global ranking of terminal capacity, so this report compiles none.

The 2025–2028 Start-Up Story: Guyana, Brazil, Saudi Arabia

Project	Operator	Start-Up	Peak (thousand bbl/d)	Notes
Guyana — Yellowtail	ExxonMobil 45% / Hess–Chevron 30% / CNOOC 25%	August 2025 (four months early)	250	FPSO “ONE GUYANA”; block output already over 900 thousand bbl/d in November 2025
Guyana — Uaru	As above	End of 2026	250	$12.7bn investment, 44 wells; Guyana's national output to exceed 1,000 thousand bbl/d after start-up
Guyana — Whiptail	As above	2027	250	Hammerhead (2029, about 150) will be the 7th project
Brazil — Búzios, multiple FPSOs	Petrobras	February 2025 to 2027	180–225 per vessel	Field output already over 1,000 thousand bbl/d in October 2025; 11 FPSOs in total before 2027
Brazil — Mero	Petrobras	October 2024, May 2025	180–212 per vessel	Field averaged 760 in Q3 2026
Brazil — Bacalhau	Equinor 40% / ExxonMobil 40% / Galp 20%	Q3 2025	220	
Kazakhstan — Tengiz Expansion (FGP)	Tengizchevroil	January 2025	+260	$48bn investment; lifted Kazakhstan's output to record levels and above its OPEC+ quota
Saudi Arabia — Marjan / Berri / Zuluf	Saudi Aramco	Q4 2025 / Q1 2026 / 2026	300 / 250 / 600	Zuluf is the largest single start-up of 2026 (estimate, ClearView)
Norway — Johan Castberg	Equinor	Q1 2025	220	Barents Sea
US Gulf of Mexico — Whale / Shenandoah / Ballymore	Shell / Beacon / Chevron	2025	100 / 100 / 75	Deepwater project cluster
Côte d'Ivoire — Baleine Phase 2	Eni	2025	40	
United Kingdom — Penguin	Shell	Q1 2025	45	A rare new North Sea start-up

Saudi Arabia's Jafurah is not an oilfield project. What started up in December 2025 is gas (target sales gas of 2 billion cubic feet per day by 2030, plus about 630,000 bbl/d of condensate and NGL). Its relevance to oil is this: it replaces about 500,000 bbl/d of crude now burned in Saudi power plants, freeing that crude for export — it must not be counted in crude output statistics.

Projects Under Construction (continued): Uganda, Suriname, Angola — and Two Still Undecided

Project	Operator	Start-Up	Peak (thousand bbl/d)	Notes
Uganda — Tilenga	TotalEnergies 56.67% / CNOOC 28.33% / UNOC 15%	First oil end of 2026	approx. 190	234 wells drilled as of June 2026
Uganda — Kingfisher	CNOOC	End of 2026	approx. 40	Central processing facility mechanically complete in August 2026, about 98% ready
Suriname — GranMorgu (Block 58)	TotalEnergies 40% / APA 40% / Staatsolie 20%	Mid-2028	220	FID October 2024, $10.5bn investment; about 50% complete in October 2026
Angola — Agogo Integrated West Hub	Azule Energy (BP / Eni)	July 2025	approx. 175–180	Incl. Ndungu (started up 2026); FPSO hull designed for 120
Angola — Begonia / CLOV Phase 3	TotalEnergies	July 2025	30 each	Tie-back to existing FPSO
Angola — Kaminho	TotalEnergies	2028	approx. 70	FID May 2024, $6bn investment, the Kwanza Basin's first large deepwater development
Iraq — Ratawi Phase 1 / West Qurna-2 Expansion	—	2026	60 each	ClearView project list
Canada — West White Rose	Cenovus	2026	80	Atlantic offshore
United States (Alaska) — Pikka	Santos / Repsol	Q2 2026	80	North Slope onshore

Two Projects Whose Status Must Be Stated Clearly

- **Namibia — Venus (PEL 56): no FID yet.** TotalEnergies (45.25%) and partners plan a final investment decision in Q4 2026, with first oil no earlier than 2029–2030, peak output of about 150,000–180,000 bbl/d and recoverable reserves of about 2 billion bbl. Namibia produced no crude in 2025 and is not a producing country. Shell's Graff project took an impairment of about $400 million in January 2025 (permeability and gas issues), with no FID date set; Galp's Mopane has been taken over as operator by TotalEnergies, with an FID target of about 2028.
- **Nigeria — Bonga South-West / Aparo: the last step before FID.** On August 24, 2026, NNPC and the contractors signed a production sharing contract addendum, described as the precursor to Nigeria's first deepwater FID since 2008; peak output of about 175,000 bbl/d, life-of-field investment of $15–21 billion. As of this issue's publication, the FID itself has not been announced. The long-running dispute over Zabazaba-Etan (potentially about 150,000 bbl/d) was resolved in March 2026, with no start-up date set.

- **UAE: no large new oilfield start-ups found for 2025–2028.** The increase comes from managing producing fields to sustain and raise capacity toward the 5 million bbl/d target; Hail & Ghasha is a gas project. This item is flagged as not verified in this round of research — which does not mean small projects do not exist.

The Licensing Map: Which Countries' Blocks Are Waiting for Money and Partners

The blocks governments put up for licensing are the public "seeking investors" list: owners and rights-holders are identified and the process has a timetable. This issue summarises the main rounds of 2025–2026 below.

Country / Round	Timing and Status	Scale and Outcome
Nigeria 2025 Round	Launched December 1, 2025; awarded	50 blocks, 143 companies submitted 200 bids; 31 companies won 37 blocks, 13 blocks received no bids; official expectation of 500 million bbl of new reserves and +300,000 bbl/d within three years
Nigeria 2026 Round	**Bid portal opens November 20, 2026**	40 blocks (onshore, shallow water, deepwater), announced at ADIPEC on November 2, 2026
Libya 2025 Round	Launched March 2025, results announced February 11, 2026	First round since 2007. Only 5 of 20 blocks awarded: O1 Eni/ QatarEnergy, O7 and C3 Repsol/TPAO consortium, M1 Aiteo, S4 Chevron; 44 applicants, 37 qualified. Target: output from 1.4 to 2.0 million bbl/d by 2030; another round to follow in 2026
Brazil Permanent Oferta Round	Awarded October 7, 2026	56 areas awarded in total (7 pre-salt production-sharing blocks + 49 concession blocks and others); signing bonus of R$530.5 million for the production-sharing blocks; contracts to be signed by February 26, 2027
Norway APA 2025 / APA 2026	2025 round awarded January 13, 2026; 2026 round adds 70 blocks	APA 2025: 57 licences to 19 companies (20 applied); APA 2026 has 21 applicants, awards expected in early 2027; the 26th ordinary round was cancelled and its acreage folded into APA
US Gulf of Mexico Lease Sales BBG1–3	December 2025, March 2026, August 2026	Each sale offers about 15,100 blocks / 80 million acres, royalty 12.5%: BBG1 high bids $300.4 million / 181 blocks, BBG2 $47 million / 25 blocks, BBG3 $82.7 million / 59 blocks; the law requires 30 sales in total before 2040

The Licensing Map (continued)

Country / Round	Timing and Status	Scale and Outcome
Egypt 2026 Round	Launched August 11, 2026; bids due November–December 2026	14 blocks (EGAS 8: Mediterranean, Nile Delta, North Sinai; EGPC 6: Gulf of Suez, Sinai, Western Desert); plus 4 Red Sea blocks; five-year plan of about 480 wells and over $5.7bn of investment
India OALP-X / OALP-XI	Round X launched February 2025; Round XI launched March 2026	Round X: 25 blocks, about 182,600 sq km; Round XI: 21 blocks, about 80,200 sq km; deepwater block bids due November 15, 2026
Indonesia 2026 Phase I	Winners announced September 17, 2026	6 blocks awarded, total investment commitment $141.51 million (Natuna D-Alpha won by Nations Petroleum at $103.3 million); Phase II bids for 8 blocks due November 2, 2026
Guyana's First Competitive Licensing Round	Bids closed September 2023; production sharing agreements signed through 2025	6 consortia selected from 14 blocks: S4 TotalEnergies/ QatarEnergy/Petronas, S8 ExxonMobil consortium and others; a further 25,000 km 2D seismic programme from 2026
Suriname Open Door	Launched November 2025, open on an ongoing basis	Covers about 60% of offshore acreage, over 70,000 sq km, replacing scheduled rounds; production sharing contracts for Blocks 9 and 10 signed at the end of 2025
Iraq Rounds 5+ and 6	Awarded May 2024, signed October 2024	14 of 29 sites awarded, Chinese companies winning the most (CNOOC, Zhongman Petroleum, ZhenHua Oil, United Energy, Anton Oil, Sinopec and others); target +750,000 bbl/d
United Kingdom	**No new licences to be issued**	The government has committed to ending new exploration licences; Round 33 (82 licences) is the last; producing fields are managed over their full life cycle
Angola Multi-Year Programme	64 blocks negotiated and 37 awarded in 2019–2025; next round expected in 2026	Next round covers the Kwanza, Benguela and Namibe basins; officials say the investment pipeline under discussion is about $70bn

Round Observations

- **Award rates diverge:** Libya's 2025 round awarded only 5 of 20 blocks (25%), while Nigeria's 2025 round awarded 37 of 50 (74%). Award rates reflect both the attractiveness of terms and current oil-price expectations and geopolitical judgement.
- **Signing has a firm deadline:** For the 56 areas awarded in Brazil in October 2026, contracts must be signed by February 26, 2027; the 6–12 months after an award are usually the most active period for farm-out and joint-development negotiations.
- **Large deepwater projects await FID:** The final investment decisions for Namibia's Venus (target Q4 2026) and Nigeria's Bonga South-West are still pending; both are the key deepwater milestones of 2026–2027.
- **The UK has exited new rounds:** The government has committed to ending new exploration licences, with Round 33 (82 licences) the last; producing fields are managed over their full lifecycle.

Part I — Upstream: Oil Production | 10. Institutions and the Landscape

After the UAE's Exit, OPEC Has Only 11 Members Left

The UAE left OPEC on May 1, 2026 (having joined in 1967), the biggest change in the institutional landscape this issue. OPEC now has 11 members: Algeria, Congo (Brazzaville), Equatorial Guinea, Gabon, Iran, Iraq, Kuwait, Libya, Nigeria, Saudi Arabia and Venezuela.

Institutional Basis	2025 Data	Notes
OPEC crude output	approx. 28.0 (incl. UAE, EIA basis)	About 35% of global crude; another compilation gives about 30, on a different definition
OPEC proved reserves	1,243.49 billion bbl (12 countries incl. UAE), 79.1% of the world	The 11 remaining members account for about 71.4%
OPEC crude exports	19.85 million bbl/d	71.9% of its 2024 exports went to Asia
Non-OPEC crude output	approx. 56 million bbl/d, about 66% of the world	The US alone accounts for 16.1% of the world

OPEC+: The Seven-Country Core and the October 2026 Production Requirements

OPEC+ (Declaration of Cooperation) comprises the OPEC members plus Russia, Kazakhstan, Mexico, Oman, Azerbaijan, Bahrain, Brunei, Malaysia, South Sudan and Sudan; the overall production arrangement has been extended to December 31, 2026. The core seven countries implementing voluntary cuts — Saudi Arabia, Russia, Iraq, Kuwait, Kazakhstan, Algeria and Oman — have been gradually restoring the earlier 2.2 million bbl/d of voluntary cuts since April 2025 (the second tranche of 1.65 million bbl/d was restored by August 2026); on September 6, 2026 they decided to hold October at September's level, and the November 1 meeting held it unchanged again.

Country	October 2026 Production Requirement (million bbl/d)	Country	October 2026 Production Requirement
Saudi Arabia	10.478	Kuwait	2.676
Russia	9.949	Kazakhstan	1.628
Iraq	4.431	Algeria	1.007
Oman	0.841	**Seven-country total**	**31.010**

Note: excl. compensation cuts. The compliance difficulty lies with Kazakhstan (Tengiz alone, at 0.93 million bbl/d, is nearly 60% of its quota, and it has long over-produced) and Iraq; the compensation period has been extended to December 31, 2026.

The IEA and Strategic Reserves: The Largest Release in History

- The IEA has 32 member countries, each required to hold stocks of at least 90 days of net imports.
- On March 11, 2026, the IEA launched the largest collective action in its history, releasing 400 million bbl in total (271.7 million bbl of government stocks + 116.6 million bbl of obligated industry stocks and

others); about 325–330 million bbl had been released by early October, and the final total will exceed 400 million bbl.

- After the release, members' public emergency stocks still stood at about 1.1 billion bbl (October 7); observable global stocks were 509 million bbl lower at the end of September than at the end of February.
- The US Strategic Petroleum Reserve (SPR) fell to 282.7 million bbl (week ending October 16), its lowest since 1982; the US pledged 172 million bbl, the largest single contribution.
- The largest buffer outside the system is in China: onshore crude stocks of about 1.19 billion bbl (week of October 19), 1.1–1.3 billion bbl including strategic underground reserves.

Part I — Upstream: Oil Production | 11. Supply, Demand and Trade

About Half the World's Crude Crosses a Border; China's Imports Hit a Record

Global crude exports were 44.5 million bbl/d in 2025, about half of world crude output. Counting refined products and re-exports as well, total oil exports were 76.59 million bbl/d, led by the US at 10.71 — but nearly two-thirds of that is refined products, which must not be confused with the crude exports in the table below.

Rank	Crude Exporter	Exports (million bbl/d)	Share of Global Crude Exports	Rank	Crude Exporter	Exports	Share
1	Saudi Arabia	6.43	14.4%	6	UAE	2.88	6.5%
2	Russia	4.54	10.2%	7	Brazil	2.06	4.6%
3	United States	3.99	9.0%	8	Norway	1.72	3.9%
4	Canada	3.67	8.2%	9	Iran	1.67	3.7%
5	Iraq	3.26	7.3%	10	Kazakhstan	1.52	3.4%

Source: OPEC Annual Statistical Bulletin 2026 compilation. US crude exports were about 4.0 million bbl/d in 2025, −3% y/y, the first decline since 2021 (−7% to Europe); US net imports fell from 2.5 to 2.2 million bbl/d over the same period (EIA).

The Import Side: China at 11.6 Million bbl/d

- **China Is the World's Largest Crude Importer**: imports of 11.6 million bbl/d in 2025 (about 578 million tonnes), a record. The top five sources on the official customs basis — Russia, Saudi Arabia, Malaysia (mostly re-exports of sanctioned crude), Iraq and Brazil — together account for 62%. Russia, at about 100.7 million tonnes (−6.9% y/y), is still first; Brazil is up 28% y/y; sanctioned crude (Iran about 1.38, Venezuela 0.389, Russia at least 0.8) totals no less than 2.6 million bbl/d, more than 22% of imports; about 430,000 bbl/d went into storage.
- Other major importers (from a single secondary compilation, not checked against customs data, indicative of scale only): the US 6.59, India 4.80, South Korea 2.76, Japan 2.32, Germany 1.69, Spain 1.30, Italy 1.13 million bbl/d.
- Seaborne volumes (January–September 2025): global loadings rose 0.6% y/y to 1,652.2 million tonnes, of which the Arab Gulf accounted for 39.7%; China's seaborne imports were 22.5% of the world total, the EU 21.2%, and India's import volumes rose 1.0% y/y.

Hormuz: The Throat of Global Trade — and the Breaking Point of 2026

Oil flows through the Strait of Hormuz were 19.87 million bbl/d in 2025 (crude and condensate 14.95, refined products 4.93), with 89% of the crude going to Asia; China, India, Japan and South Korea together accounted for 74%. By exporting country, flows through the Strait were Saudi Arabia 6.23, Iraq 3.63, UAE 3.24, Iran 2.41 and Kuwait 2.37 million bbl/d. Bypass capacity is far from enough: Saudi Arabia's East–West pipeline has a nominal capacity of 7 million bbl/d (about 5 million usable for exports) and the

UAE's Habshan–Fujairah pipeline 1.8 million bbl/d; the IEA estimates effective bypass at only 3.5–5.5 million bbl/d in total. After military action on February 28, 2026 and the Strait's closure on March 2, vessel transits fell by about 95% at one point in March, flows dropped to 4.9 million bbl/d in the second quarter, and partially recovered to 10.9 million bbl/d in September (Gulf crude exports including pipelines were 13.9 million bbl/d, more than 75% of the pre-war level).

Prices Are High, yet Investment Is Falling — This Issue's Most Important Contrast

- **US shale and the Permian.** Permian output rose from 6.3 to 6.6 million bbl/d in 2025, 48% of US crude output. Oil rigs recovered from about 405–410 at the start of 2026 to 462–463 in October–November; but the Dallas Fed's Q3 E&P capital expenditure index fell from 42.7 to 35.3 — companies still plan on a $65–70 budget price, and capital discipline outweighs high spot prices.
- **South America's big three.** Guyana averaged 750,000 bbl/d in 2025 and passed 900,000 by year-end; after Uaru starts up it could pass 1 million by end-2026, with targets of 1.55 million in 2029 and over 1.7 million in 2030. Brazil's crude output hit a record of about 4.0 million bbl/d in October 2025, about 80% of it pre-salt; Argentina's Vaca Muerta shale oil exceeded 447,000 bbl/d. JPMorgan estimates Brazil, Guyana and Argentina together could add a further 750,000–1 million bbl/d of non-OPEC+ supply.
- **The sanctions landscape.** Russia's 2025 output was flat at 9.89 million bbl/d; after the US sanctioned Rosneft and LUKOIL in October 2025, its exports to China fell but it remained China's top supplier. Iran's output rose from 2.51 million bbl/d in 2020 to 4.05 million in 2025, absorbed mainly by China's independent refineries (about 1.38 million bbl/d); Venezuela's output recovered to 970,000 bbl/d (+13%).
- **Investment and transition.** IEA World Energy Investment 2026: global energy investment in 2026 is about $3.4 trillion (+5%), of which clean energy about $2.2 trillion and fossil fuels about $1.2 trillion; **upstream oil investment fell for a third consecutive year, to below $500 billion in 2026**, despite high oil prices — the IEA attributes this to doubts over whether high prices will last, long project cycles and supply-chain constraints. US shale spending is expected to fall 7% in 2026; about 75% of 2026 investment was already locked in before the war. Upstream investment in Africa was $37 billion in 2025, far below the $68 billion of 2016.
- **Divergent demand outlooks (forecasts, not observations).** Before the war the IEA expected demand to grow by about 700,000 bbl/d in each of 2025 and 2026; after the war its September 2026 report put 2026 demand at −2.5 million bbl/d; the EIA, meanwhile, expects demand to recover to 104.64 million bbl/d in 2027, with Brent falling back to about $84. The three figures come from different points in time and cannot be compared side by side.

The Competitive Landscape: Who Is Competing for Field Owners

Type of Competitor	Approach
International oil companies (ExxonMobil, Chevron, Shell, TotalEnergies, BP)	Win deepwater and large blocks with operating capability and balance-sheet strength; Guyana, Brazil's pre-salt and Suriname are all in their hands
National oil companies and sovereign capital	Lock in resources through state-to-state cooperation and long-term offtake; Chinese companies winning the most in the Iraq rounds is a case in point

Independent small and mid-sized oil companies (Perenco, Azule and others)	Specialise in mature and marginal fields, making money from low-cost operations and fast production gains; Perenco in Chad is an example
Commodity traders and prepayment financing	Provide financing against crude prepayment and offtake — fast, but on tough terms
Other RWA / tokenisation platforms	Few comparable oil-asset tokenisation projects are publicly visible, and there is no scaled precedent to benchmark against

Chad: Two Authoritative Sources Disagree, So This Report Lists Both

The output figures conflict — stated up front. EIA basis: Chad's 2025 crude output was 126,490 bbl/d (−0.6% y/y); Wood Mackenzie basis: about 190,000 bbl/d. The gap may come from different definitions of crude versus all liquids, or of operated output; this round of research could not resolve it. This report headlines the EIA figure; anyone citing Chad asset material must check which basis the other side uses — the first item to align when doing due diligence on MoW001 (Chad Doba) issuance material.

Item	Content
Proved Reserves	1.5 billion bbl (2025), about 0.08% of the world; R/P ratio about 32 years on EIA output
Five-Year Output Series (EIA, 10,000 bbl/d)	2020: 12 • 2021: 11 • 2022: 11 • 2023: 12 • 2024: 13 • 2025: 13
History	The Doba project started up in 2003 and peaked within a year at 183,000 bbl/d; by 2013 it had fallen to 83,000 bbl/d; Benoy, Block H, Badila and Mangara then lifted it back up. Wood Mackenzie judges that without new discoveries, output stays roughly flat until 2027 and then declines slowly
Main Basins and Fields	Doba Basin (south): the Komé–Miandoum–Bolobo complex, about 52.4% of Chad's upstream market value in 2025 (market research estimate, secondary source); Bongor Basin (operated by CNPC, supplying the Djermaya refinery at 20,000 bbl/d; CNPC 60% / government 40%); Badila and Mangara (Perenco)
Operator Developments	Doba was originally developed by an ExxonMobil-led consortium (World Bank financing, 2003); the assets were transferred by decree to the national oil company SHT in March 2023. After Perenco (PetroChad Mangara) worked over 12 wells, Badila/Mangara capacity exceeded 18,000 bbl/d (December 2025), with peak additions of 7,000 bbl/d including 8 horizontal wells; the rig has moved to the Krim development in the Doba Basin (8 wells planned). The Kombi 2 gas project is expected to start up in early 2026, adding about 10 million bbl of reserves
Export Route	The 1,070 km Chad–Cameroon pipeline to the Kribi terminal in Cameroon, in service since 2003, designed for about 225,000 bbl/d; secondary sources put pipeline exports at about 110,000 bbl/d, broadly consistent with EIA output. Doba crude is heavy and high-acid, traded at a discount to Brent

Chad's asset logic is a mature basin, low-cost production growth and a single export route: output flexibility comes from Perenco-style infill drilling and workovers on old wells, not new discoveries. Cash flow from Doba-type assets depends on three variables — the decline curves of producing wells, the start-up pace of replacement projects such as Krim, and the actual realised price at Kribi loadings and discounts. Reserves of 1.5 billion bbl and an R/P ratio of about 32 years show the resource base is long enough; the risks lie mainly in operational continuity and the export route, not in resource exhaustion.

Canada: Oil Sands Are 70% of Output; Cold Lake Is the Thermal-Recovery Example

Item	Data and Basis
Crude output (EIA)	4.96 million bbl/d (2025, +3.9%), 4th in the world, 5.9% share
Crude and equivalents (Canada Energy Regulator, CER)	A record 5.35 million bbl/d (2025), peaking at 5.64 in December; Alberta accounts for 83.8% of the national total. The difference from EIA is statistical scope (incl. equivalents/NGL)
Oil sands output	About 3.5 million bbl/d (CER); the industry association CAPP's basis (mining + in situ) exceeds 3.7, on a different definition. In situ thermal capacity hit a record 1.67 million bbl/d in 2025, a fifth consecutive annual high. Conventional (non-oil-sands) output is about 1.5 — the remainder of 4.96 minus 3.5 — this report's own estimate, not an official split
Proved Reserves	About 170 billion bbl in total (Natural Resources Canada, Energy Fact Book 2025–26), about 97% in oil sands (about 165 billion bbl); the Alberta regulator's basis gives proved + probable bitumen reserves of 187 billion bbl (end-2023, incl. probable, not directly comparable)
The three oil sands regions	Athabasca, Cold Lake and Peace River, all in Alberta, covering about 142,000 sq km; the Pathways Alliance's carbon-capture hub is planned near Cold Lake
Rigs	223–226 (late October to early November 2026), of which 154 oil rigs
Five-Year Output Series (EIA, million bbl/d)	2020: 4.18 • 2021: 4.44 • 2022: 4.54 • 2023: 4.59 • 2024: 4.78 • 2025: 4.96 — no sharp moves, steady year-by-year growth

Cold Lake (Imperial Oil)

- **Output: gross production of 149,000 bbl/d in Q2 2026** (145,000 in the same period of 2025); 152,000 bbl/d in the first half of 2026 (150,000 in the same period last year); Cold Lake blended crude sales were 195,000 bbl/d in the same period. Source: Imperial Oil Q2 2026 results announcement, July 31, 2026.
- The process is cyclic steam stimulation (CSS) and solvent-assisted SAGD; CAPP industry material lists Imperial Oil's Cold Lake CSS bitumen capacity at about 190,000 bbl/d (calendar-day basis, undated) — the capacity basis differs from the actual gross output basis, and the two must be kept apart when cited.
- The top three in situ thermal producers in 2025: Firebag (Suncor) 245,000, Foster Creek (Cenovus) 206,000 and Surmont (ConocoPhillips) 146,000 bbl/d. Cold Lake is on a par with Surmont — a large second-tier thermal asset.

Canada and Chad are two very different asset types: oil sands thermal recovery declines slowly, lasts long and is highly predictable, at the cost of high energy and carbon costs and expansion that depends on large capital spending; cash flow from assets like Cold Lake looks more like a long, gentle curve. When assessing such assets, the key variables are the steam-to-oil ratio, gas costs and the carbon price — not geological risk.

Part II — Downstream: The Oil Industry | 14. Oil Consumption Structure

Of Every Hundred Barrels of Oil, About Fifty-Eight Are Burned on the Road

Part II covers macro statistics and analysis only, answering one question: who uses the oil, and for what? Start with the total — IEA Global Energy Review 2025: global oil demand was 193 exajoules in 2024, +0.8% y/y; oil's share of all energy demand fell below 30% for the first time (its peak, about 50 years ago, was 46%). Demand in 2024 was only 1.3% above 2019, with almost all of the increase coming from petrochemical feedstock (+12% over five years).

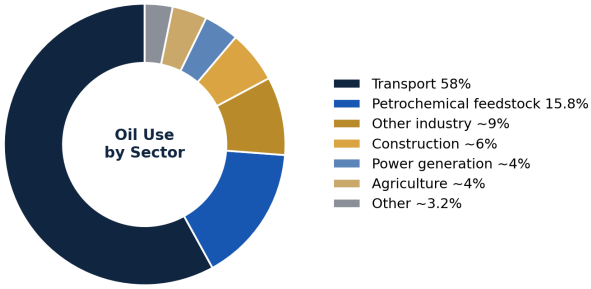


Figure 3 • Global oil consumption by use. The IEA does not publish a single sectoral table in million bbl/d: transport at 58% is the IEA World Energy Outlook 2025 energy basis (112/193 EJ), petrochemical feedstock at 15.8% is the IEA Oil 2025 volume basis (2024); the remaining shares are indicative figures from secondary compilations, and the volumes in the table below are those shares multiplied by total demand of about 103 million bbl/d, as labelled.

Sector	Share	Estimated Volume (million bbl/d)	Basis
Transport, total	approx. 58%	approx. 59–60	IEA WEO 2025 energy basis; OPEC basis over 57% in 2024
Of which: road transport	approx. 45%	approx. 46	IEA
Of which: aviation	approx. 7%	approx. 7–8	cross-checks with jet fuel demand of 7.76 (Chapter 15)
Of which: shipping	approx. 5–6%	approx. 5–6	Bunker fuel demand approx. 5 (Chapter 15)
Petrochemical feedstock (non-energy use)	15.8%	approx. 16.3	IEA Oil 2025, volume basis (2024); rising to 17.4% and 18.4 million bbl/d by 2030
Other industry (non-feedstock)	approx. 9%	approx. 9	Secondary compilation; indicative
Construction (heating oil, LPG)	approx. 6%	approx. 6	Secondary compilation; weak sources; indicative
Power generation	approx. 3–5%	approx. 3–5	Mainly in the Middle East; total fuel oil demand of 6.5 includes shipping and industry and cannot all be counted as power generation

Agriculture	approx. 4%	approx. 4	Secondary compilation; no standalone official figure for agricultural diesel (Chapter 19)
Other (residual)	approx. 5%	approx. 5	Residual balancing item

The growth structure tells more than the stock structure: since 2022, road transport has contributed only about 5% of oil demand growth; from 2019 to 2024, petrochemical feedstock added 2.3 million bbl/d, more than 95% of net demand growth; non-feedstock oil consumption in 2024 was roughly flat against 2019, even though global GDP grew about 14% over the same period. Oil demand has not disappeared — its growth point has simply moved from the fuel tank to the chemical plant.

Part II — Downstream: The Oil Industry | 15. Transport Fuels

Diesel Is the Largest Single Product; Gasoline's Rival Is the Electric Vehicle

Product (thousand bbl/d)	2019	2024	2025	2026 forecast*
Motor gasoline	26,928	27,452	27,788	27,908
Diesel (gas oil)	28,747	28,661	28,926	29,053
Jet fuel (jet fuel and kerosene)	7,865	7,534	7,763	7,879
LPG and ethane	13,211	14,985	15,250	15,403
Naphtha	6,690	7,192	7,191	7,403
Fuel oil (residual fuel oil)	—	approx. 6,200	approx. 6,500	approx. 6,100 in 2030 (forecast)

Source: IEA Oil Market Report, March 2026 (the last pre-war edition), global demand table by product; fuel oil from IEA Oil 2025. *The 2026 forecast follows the March edition's basis; the May–June 2026 reports revised full-year demand down sharply because of the war (about −1.1 million bbl/d). The two versions must not be mixed.

Four Trends

- **EV displacement.** IEA Global EV Outlook: EVs displaced more than 1.3 million bbl/d of oil in 2024 and about 1.7 million bbl/d in 2025 (of which cars alone about 1.2 million, and China about 1.0 million), projected to exceed 5 million bbl/d by 2030 (China about 2.7 million). EV sales in 2025 exceeded 20 million units, about a quarter of new-car sales. To put it in perspective: the 2025 displacement is roughly equal to Japan's entire transport oil use.
- **China's gasoline is already on a downward path.** China's gasoline demand was 3.65 million bbl/d in 2024, roughly flat in 2025, and is forecast at 3.53 million in 2026 (−3.3%); but China's jet fuel is still growing (0.94 → 0.97 → 1.01 million bbl/d).
- **The diesel plateau.** Diesel is the largest single product (about 29 million bbl/d); road freight is the main user, with heavy commercial vehicles accounting for about 60% of road diesel; most institutions expect it to peak and turn down in the early 2030s.
- **Bunker fuel neither rises nor falls.** Marine fuel demand is about 5 million bbl/d, and the IEA expects it to stay roughly flat from 2024 to 2030; in 2024, rerouting around the Cape of Good Hope because of the Red Sea added about 140,000 bbl/d. Singapore is the world's largest bunkering port: sales of 54.9 million tonnes in 2024 and a record 56.77 million tonnes in 2025. Total fuel oil demand in 2025 was 6.5 million bbl/d, still 4.8% above 2019 — held up by the shadow fleet and by Middle Eastern power plants burning oil.

What this means for you: Gasoline sets the pace at which the existing fleet is replaced; diesel and jet fuel set the floor under freight and travel; bunker fuel sets the cost of global trade. Diesel, jet fuel and bunker fuel together are nearly 42 million bbl/d — the hardest part of global crude demand to replace, and the physical anchor of the long-term oil price.

Petrochemicals: The Industry Where Oil Stops Being Burned and Starts Staying

The global petrochemicals market in 2025 is put at anywhere from $582 billion (NAVADHI) to $698 billion (Cervicorn), depending on definition, with IMARC at $675.7 billion. This report cites the range of $580–700 billion and takes no single point. The ethylene market alone is about $219.3 billion (2025).

Product	Global scale	Notes
Ethylene	Capacity approx. 235 million t/yr; demand approx. 190 million t (2025)	Utilisation about 80%, the weakest in 40 years, driven mainly by capacity expansion in China; Sinopec's ethylene output hit a record 13.47 million tonnes in 2024
Ethylene capacity distribution	US approx. 45–46 million t/yr; China approx. 36.7 million t in 2024, reaching approx. 51.8 million t by 2030	The centre of capacity keeps moving to China
Propylene	Output approx. 135–150 million t (2025 estimate)	About 65% goes into polypropylene
Plastics	Output 430.9 million t (2024, Plastics Europe)	Up 4.1% y/y; primary-form plastics output value approx. $998 billion (IndexBox, wider basis); polypropylene 80 million t, polyethylene 76 million t, PVC 40 million t; packaging accounts for 36.5% of plastics use (2023)
Synthetic rubber	Output approx. 16.3 million t (2024, IRSG basis, 52% of all rubber)	IndexBox's wider basis gives 25 million t; the two bases conflict and are shown side by side

Chemical Fibres: 69% of the Fibres We Wear Are Synthetic

Fibre	2024 output	Share of all fibres	Notes
All fibres	132 million t	100%	Up 5.6% y/y; projected to reach 169 million t by 2030
Synthetic fibres, total	91 million t	69%	China accounts for about 65% of synthetic fibre output
Polyester	77.7 million t	59%	The single largest fibre; of which recycled polyester 9.3 million t; China accounts for about 60% of polyester output
Nylon (polyamide)	approx. 7 million t	approx. 5%	
Other synthetics (polypropylene fibre, acrylic, spandex, etc.)	approx. 6.6 million t	approx. 5%	Residual approximate figure
Cotton (for comparison)		19%	

	24.5 million t	The largest natural fibre, yet only a third of polyester's output

Source: Textile Exchange, Materials Market Report 2025. The polyester market is worth about $111.6 billion (2025, secondary source, indicative).

What this means for you: Petrochemical feedstock is the only major part of oil demand still clearly growing — clothes, packaging, auto parts and medical supplies all come from here. Transport electrification squeezes gasoline, not chemical fibres and plastics — and that is the key to understanding why 'demand peaking' does not mean 'crude value peaking'.

First, Correct a Common Misconception: Fertilisers Run Mainly on Natural Gas, Not Oil

Ammonia is made from hydrogen plus nitrogen: today the hydrogen comes mainly from natural gas steam reforming (in China, partly from coal), and urea is ammonia plus carbon dioxide. Oil's link to fertilisers runs through three indirect lines: first, a minority of plants use naphtha or heavy oil as feedstock; second, gas prices move with oil prices over the long term, so a gas price shock (for example the 2026 Hormuz disruption) pushes ammonia and urea prices up directly; third, large integrated petrochemical complexes in the Gulf and China produce olefins and ammonia/urea together. Calling fertilisers an 'oil product' is inaccurate; this report treats them as an energy product.

Product	2024 scale	Sources and conflicting definitions noted
Ammonia output	approx. 190.5 million t	International Fertilizer Association (IFA); capacity approx. 240 million t, about a tenth traded
Urea output	approx. 201 million t	IFA basis, up 3% y/y; IndexBox basis 143–146 million t — different statistical coverage; the conflict is shown side by side
Fertiliser consumption (nutrients N + P_2O_5 + K_2O)	approx. 192 million t of nutrients	IFA / FAO compilation; nitrogen fertiliser consumption about 115 million t of nitrogen, urea about 46% of nitrogen fertiliser by volume
Fertiliser output (all products)	approx. 614 million t of products	IndexBox; top three producers: China 97 million t, US 66 million t, Russia 64 million t
Fertiliser market size	$214 billion (nutrient basis) / $299.2 billion (product basis)	Both bases shown side by side; no average taken
Urea market	Consumption 157 million t, about $62.8 billion	IndexBox

Pesticides: A Range of $85–130 Billion

For the 2025 global pesticides market, institutions give widely differing figures — $84.6 billion (Dataintelo), $108.4 billion (Reports and Data), $115.1 billion, $128.2 billion (IndexBox) — so this report cites only the $85–130 billion range and names the source whenever a single figure is quoted. Herbicides are the largest category, about 38–41%. The main producers are Syngenta, Bayer, BASF, Corteva and FMC. The synthesis chain for pesticide active ingredients runs heavily through petrochemical intermediates (aromatics, chlor-alkali and solvents) — it is the petrochemical downstream's highest-priced, lowest-volume branch.

Cost-structure note: Natural gas accounts for about 70–90% of ammonia's cash cost (an industry rule of thumb, not a statistically measured value; this report flags it as an estimate). That means gas prices — and with them the oil prices and shipping routes linked to gas — directly set fertiliser prices and food costs. The fertiliser price swings of 2026 are, at bottom, Hormuz transmitted downstream.

A $1.7 Trillion Pharmaceuticals Market — but No Authority Can Put a Number on the Oil-Derived Share

The global pharmaceuticals market was about $1.67–1.74 trillion in 2024 (DataM Intelligence $1.7404 trillion; Beyond Market Insights $1.6677 trillion) and about $1.77 trillion in 2025. Petrochemicals connect with pharmaceuticals at three points: solvents, the intermediates for the active ingredients of a large number of small-molecule drugs (benzene, toluene, xylene, and ethylene and propylene derivatives), and excipients and plastic packaging. But no authoritative source has ever quantified 'oil-derived pharmaceutical output' on its own — so this report describes only the dependence and does not invent that share.

- Small molecules (traditional chemical drugs) account for about 54% of the pharmaceuticals market and are the part most dependent on petrochemical feedstocks; biologics have risen to about half (IQVIA), and their production depends far less on petrochemical feedstocks.
- The synthesis routes for common drugs such as aspirin and ibuprofen all pass through benzene-based intermediates; the disposable IV bags, syringes and medicine bottles in hospitals come directly from polypropylene and polyethylene — the link between pharmaceuticals and petrochemicals lies more in 'how each pill is made and packaged' than in what the drug's price is made of.
- The citation rule is therefore fixed: pharmaceuticals are 'a $1.7 trillion downstream market partly dependent on petrochemical inputs' — never 'an oil-derived pharmaceuticals market worth $1.7 trillion'.

Daily-Use Chemicals: Surfactants Are the Main Battlefield

Market	Scale	Link to petrochemicals
Detergent chemicals	approx. $67.3–71.4 billion (2024–2025)	Surfactants account for about 32%, the largest category; their feedstocks split into two lines — the petrochemical line (linear alkylbenzene, from kerosene and benzene) and the oleochemical line (palm, coconut), with bio-based substitution advancing. Narrow-basis institutions give $43.5 billion, shown alongside for reference
Cosmetics and beauty	approx. $336 billion (2024)	Secondary compilation; weak sources; magnitude only: solvents, emollients, polymers and packaging all have petrochemical sources
Care chemicals (surfactants, solvents and additives for home and personal care)	$11.25 billion (2024)	Narrow-basis definition; projected to reach $17 billion by 2032
Solvents (standalone global market)	**Unverified**	Published figures mostly mix coatings, pharmaceuticals, cleaning and other uses, and their bases are incompatible; this issue cites no single figure

What this means for you: Pharmaceuticals and daily-use chemicals are small in volume within crude demand, yet they are the most direct evidence of 'oil entering everyday life' — pills, shampoo, laundry liquid, cosmetics and their packaging all sit on the same petrochemical chain. Reading the downstream means looking not only at the oil that gets burned, but also at the oil that stays behind as things.

Food Meets Oil at Three Points: In the Field, on the Road, and in the Packaging

Item	Scale	Notes
Food and beverage plastic packaging	$45.67 billion (2024)	Reports and Data; projected at $78.12 billion by 2034; rigid plastics about 60%
Fresh food packaging	$134.6 billion (2024)	Secondary source; indicative
Chilled food packaging	$38.24 billion (2025)	Of which plastics 53% (about $20.3 billion)
Cold chain market (all bases)	$368.1–405 billion (2024)	Definitions vary widely: logistics-only basis about $260 billion; food and beverage is the largest application, pharmaceuticals the fastest-growing
Packaging's place within plastics	36.5% of global plastics use (2023)	Plastic packaging waste in 2024 about 156 million t (OECD compilation)
Agricultural diesel use	**No public global million-bbl/d figure**	Only the statement that agriculture accounts for about 4% of oil consumption (implying about 4 million bbl/d, not measured); this report does not cite estimates as statistics

- **In the field:** Diesel for tractors and harvesters, fertilisers (the natural gas chain of Chapter 17), pesticides (petrochemical intermediates) — modern agriculture's yields rest on fossil energy inputs; the FAO notes that insufficient refrigeration causes about 526 million tonnes of food loss a year (March 2026), and the cold chain itself is a heavy user of diesel and electricity.
- **On the road:** Food is one of the largest categories in global freight, and diesel prices feed directly into food's landed cost; the first consumer prices the high oil prices of 2026 reached were freight and fertilisers.
- **In the packaging:** Food packaging is the largest use of plastics, and resin accounts for about 50–70% of packaging cost (an industry rule of thumb, an estimate) — oil price moves do eventually show up in the packaging cost on supermarket shelves, only diluted layer by layer through processing and distribution.

What this means for you: The point of the food chapter is not the money, but the chain — from sowing, fertilising, harvesting, refrigeration and transport to packaging, a piece of food meets the oil industry chain at least four times before it reaches the table. That is also why the oil price has never been only a motorist's concern, but everyone's price concern.

Part II — Downstream: The Oil Industry | 20. Industry Output Value in Summary

Put the Whole Chain's Money in One Table — but Read the Basis Before the Size

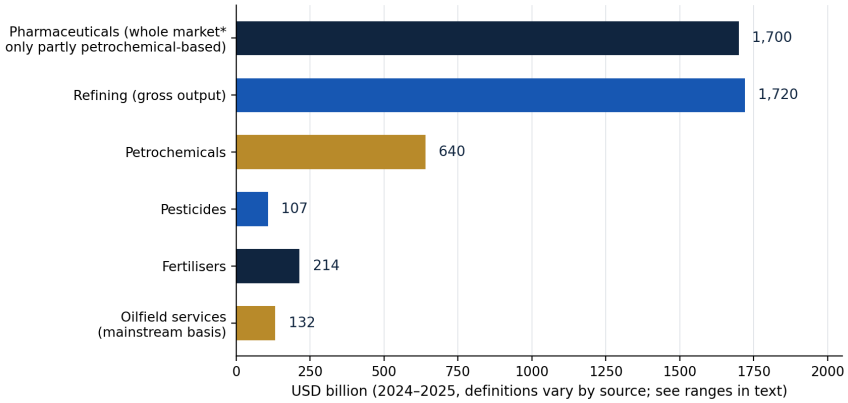


Figure 4 • Output value by major industry (USD billion; the chart shows range midpoints or representative values — see the table below for the exact ranges). *Pharmaceuticals is the whole-market size and only partly depends on petrochemical feedstocks; it must not be counted as oil-derived output value.

Industry	Scale (latest)	Basis
Refining (gross output)	$1.61–1.84 trillion (2025)	Expert Market Research $1.611 trillion, IMARC $1.674 trillion, Reports and Data $1.842 trillion; this is mostly crude value passing through — refinery value added (a $5–20/bbl margin against $60–100 oil) is only about 5–15%. The narrow basis counting only services and processing fees is just $67–143 billion
Pharmaceuticals (whole market, for comparison)	$1.67–1.74 trillion (2024)	Only partly dependent on petrochemical inputs; no authoritative quantification of the oil-derived share — listed only as a downstream scale comparison
Petrochemicals	$582–698 billion (2025)	A definition-driven range; see Chapter 16
Fertilisers	$214 billion (nutrient basis) – $299.2 billion (product basis, 2024)	Main feedstock is natural gas; see Chapter 17
Oilfield services	$119–144 billion (mainstream basis, 2024–2025)	The wide basis can reach $320–350 billion; see Chapter 6
Pesticides	$85–130 billion (2025)	Large differences between institutions; see Chapter 17
Food and beverage plastic packaging	$45.67 billion (2024)	For the full packaging basis, see Chapter 19

How Much of Each Link's Cost Is Crude Oil

No public source offers an authoritative 'oil cost share by industry' table. The only figures that hold up are rules of thumb, which this report flags as estimates: crude accounts for about 85–90% of variable cost and output value in refining; naphtha, ethane, LPG and other feedstocks account for about 60–80% of ethylene crackers' cash cost; natural gas accounts for about 70–90% of ammonia's cash cost; resin

accounts for about 50–70% of plastic packaging cost. These figures are for estimating pass-through elasticity only — they are not statistical facts and must not be cited as measured data.

Four Things the Downstream Section Says to Mo Users

1. Crude's Value Is Anchored in Real Consumption, Not Speculation

The world consumes about 104 million bbl/d of liquid fuels and feedstocks: about 58% for transport, 15.8% turned into feedstocks for plastics, chemical fibres, rubber and pharmaceuticals, and the rest into industry, construction, power generation and agriculture. This is not demand that can disappear overnight — EVs displaced about 1.7 million bbl/d in 2025, only about 1.6% of total demand, and the displacement is concentrated in passenger-car gasoline; diesel freight, aviation, shipping and petrochemical feedstocks together exceed 50 million bbl/d and still have no large-scale substitute. The value of oilfield crude rests on exactly this consumption list.

2. Demand Growth Has Moved — and That Decides Which Oilfields Are Worth More

More than 95% of net oil demand growth since 2019 has come from petrochemical feedstocks. Petrochemical plants want stable, long-term, predictable crude and condensate supply, and prefer light, low-sulphur crude. The implication for oilfield assets is direct: fields with stable output, slow decline and the ability to sign long-term sales contracts fit the downstream's new demand structure better than fields with high but volatile output — the same logic as Chapter 13's analysis of Chad (raising output from mature fields) and Canada's oil sands (long, flat production curves).

3. Most of the Chain's Money Is Pass-Through; Real Value Added Sits in a Few Links

Of refining's $1.6–1.8 trillion of gross output, about 85–90% is just crude value passing through; oilfield services' $120–140 billion is earned on technology and equipment; petrochemicals' roughly $600 billion is earned by turning feedstocks into materials. Ordinary users used to have only one relationship with this chain: paying to consume. Mo sits at the start of the chain — oilfield crude itself: anchored in crude oil, it turns oilfield crude into digital certificates, letting users worldwide take part in the allocation of oilfield crude. For the first time, users stand on the other side of paying.

4. Impact on the Industry: Connecting Oilfield Owners Directly with Global Participants

The upstream licensing map shows that every year a large number of blocks and mature oilfields are looking for capital and partners, while upstream investment in 2026 shrank for a third consecutive year, to below $500 billion — owners short of money and users short of access are two sides of the same gap. As Mo keeps listing more oilfield assets and develops into the world's leading Web3.0 RWA petroleum asset platform, its industry significance is to give small and medium oilfield owners one more channel that does not depend on traditional financing, and to give every ordinary person on the downstream

consumption list one more place at the source. The continuing coverage of this report (the Monthly), the Mo Oil Industry Weekly Report and the Mo News Brief is to keep updating this panorama month by month, so that participants judge each new oilfield by data rather than by narrative.

Appendix I — Data Notes and Key Sources

Data Notes

- **Timing.** The research for this issue was completed on November 8–9, 2026. The latest complete year is 2025; 2026 figures are year-to-date or monthly estimates, revised month by month as the Hormuz disruption unfolds. Anything marked 'forecast' is an institutional forecast, not an observed value.
- **Basis priority.** Production rankings use the EIA crude (including lease condensate) basis; reserves use the OPEC Annual Statistical Bulletin basis, with the Eni basis also given; prices use EIA spot; rigs use Baker Hughes; refining capacity uses the EI Statistical Review. Wherever another basis is used, it is noted table by table.
- **Conversions:** Million tonnes/year are converted at 7.33 bbl/t, with an error of about ±5%; the R/P ratio is this report's static value calculated on that year's production; Saipem's euro revenue is not converted into dollars, and the currency is stated in the original.
- **Grading.** Primary sources and authoritative compilations (EIA, IEA, OPEC, EI, Baker Hughes, company announcements) are the main evidence in the text; market sizes from commercial research firms are always given as ranges; figures from secondary compilations and weaker sources are flagged in the text wherever they appear as 'secondary' or 'indicative'.

Key Sources (selection, all consulted November 8–9, 2026)

EIA International Energy Statistics (compiled via StatsGeo)https://statsgeo.com/crude-oil-production-by-country • EIA STEO, October 2026, Table 3a https://www.eia.gov/outlooks/steo/tables/pdf/3atab.pdf • EIA producing wells statistics https://www.eia.gov/petroleum/wells/ • EIA prices https://usoilgauge.com/oil-prices/history • EIA/AtlasBig 99-country production table https://uk.atlasbig.com/energy/oil-production-by-country • OPEC Annual Statistical Bulletin 2026https://publications.opec.org/asb/about/2903 • OPEC reserves table (via BusinessStats)https://businesstats.com/share-of-oil-reserves-of-the-leading-ten-countries/ • OPEC press release, September 6, 2026 https://www.opec.org/pr-detail/1835613-6-september-2026.html • EI Statistical Review of World Energy 2026, refining data (via Energy Factbook)https://energyfactbook.com/world/refining/ • IEA Global Energy Review 2025 — Oilhttps://www.iea.org/reports/global-energy-review-2025/oil • IEA Oil Market Report, March 2026 https://iea.blob.core.windows.net/assets/a25ddf53-cd6c-4910-ac90-16bfd28399e7/-12MAR2026_OilMarketReport.pdf • IEA Global EV Outlook 2025https://www.iea.org/reports/global-ev-outlook-2025/outlook-for-energy-demand • Baker Hughes rig counts (via OGJ)https://www.ogj.com/drilling-production/news/55415736/canada-rig-count-increases-us-count-holds • Columbia University Center on Global Energy Policy, 'China's Crude Imports 2025'https://www.energypolicy.columbia.edu/where-china-gets-its-oil-crude-imports-in-2025-reveal-stockpiling-and-changing-fortunes-of-certain-suppliers-including-those-sanctioned/ • Oilfield services market https://www.mordorintelligence.com/industry-reports/global-oil-field-services-market-outlook-industry • Projects under construction list (ClearView, June 2026)https://mcusercontent.com/0554cc7ed0bda904329a48c93/files/c42057ea-389c-1cbc-39a3-a25c0358e294/2026_06_15_Oil_Projects.pdf • Libya licensing round (OGJ)https://www.ogj.com/exploration-development/news/55356829/global-oil-and-gas-companies-awarded-exploration-blocks-in-libya • Nigeria 2026 round https://crispng.com/nuprc-2026-licensing-round-40-oil-blocks-bids-november-20/ • Brazil round https://www.offshore-mag.com/regional-reports/latin-america/news/55410631/brazils-anp-awards-56-exploration-areas-in-dual-oct-7-rounds • Imperial Oil Cold Lake results https://news.imperialoil.ca/news-releases/news-releases/2026/Imperial-announces-second-quarter-2026-financial-and-operating-results/default.aspx • Chad (Wood Mackenzie summary)https://www.woodmac.com/reports/upstream-oil-and-gas-chad-upstream-summary-957451/ • Fibres (Textile Exchange, via TexData)https://www.texdata.com/news/Yarn&Fiber;/22094.html • Petrochemicals market https://www.cervicornconsulting.com/petrochemicals-market • Fertilisers (IndexBox)https://www.indexbox.io/blog/fertilizer-world-market-overview-2024-13/

Appendix II — Conflicting Definitions and the No-Public-Data List

Conflicting Definitions Not Resolved in This Issue (quote them together with the conflict)

Item	Conflict
Global crude production	EIA 84.38 vs OPEC 74.85 million bbl/d — different statistical coverage
Global proved reserves	OPEC 1,572.46 vs Eni 1,802.5 billion bbl — the difference is mainly Canada's oil sands
Chad 2025 production	EIA 126,490 bbl/d vs Wood Mackenzie approx. 190,000 bbl/d — unresolved
Oilfield services market size	Mainstream basis $119–144 billion vs wide basis $320–350 billion
Refining market size	Gross-output basis $1.61–1.84 trillion vs services basis $67–143 billion
Petrochemicals / pesticides / fertilisers markets	Petrochemicals $582 billion vs $698 billion; pesticides $85 billion vs $128.2 billion; fertilisers $214 billion vs $299.2 billion
Urea output / synthetic rubber	Urea: IFA approx. 201 million t vs IndexBox 143–146 million t; synthetic rubber: IRSG 16.3 million t vs IndexBox 25 million t
EACOP capacity	216 vs 246 thousand bbl/d — two figures given
Libya round size	20 vs 22 blocks offered — two figures given (5 awarded, undisputed)
Saudi oil rig count	66 (YCharts) vs 118 (Eulerpool) — different bases
Offshore share of production	IEA approx. 30% (crude) vs Rystad about half (oil and gas equivalent) — different units
IEA demand versions	March 2026 pre-war edition vs June post-war revision (2026 demand −1.1 million bbl/d); this report's by-product table uses the March edition and says so

No Public Data (this report does not estimate or invent)

- Producing wells: apart from the United States (2024) and the six countries in the 2014 OGJ snapshot, no other country publishes comparable current figures.
- Field-level 2025 actual production: for Prudhoe Bay (current output), Safaniya, Khurais, Shaybah, West Qurna's individual fields, Lula/Tupi and Johan Sverdrup, only capacity figures or old figures exist.
- OPEC members' actual production for October–November 2026 (against quota): only the required-production table exists; actual compliance figures were not obtained.
- Baker Hughes full country-by-country rig table for September–October 2026: inside its paywalled portal; not obtained through public channels.
- Ranking of global crude export terminal capacity: no unified comparable table exists, only figures by individual port.
- Global agricultural diesel use, the standalone solvents market size, the oil-derived share of pharmaceuticals (in monetary terms), and an authoritative oil cost-share table by downstream industry: none of these were found — this report gives only the flagged share statements and rules of thumb.
- Petrofac 2024–2025 revenue, and the UAE's new oilfield start-up projects for 2025–2028: no reliable source was verified in this round.